DESIGN REVIEW · FOR FEEDBACK

Sienna Camper Conversion

A modular, tool-free lift-out sleeping + galley system for a Toyota Sienna — sleeps two, with a slide-out fridge and camp kitchen, EcoFlow power and A/C, and no permanent modification to the van. This page summarizes every feature and shows the drawings, so you can skim it and flag anything you'd change.

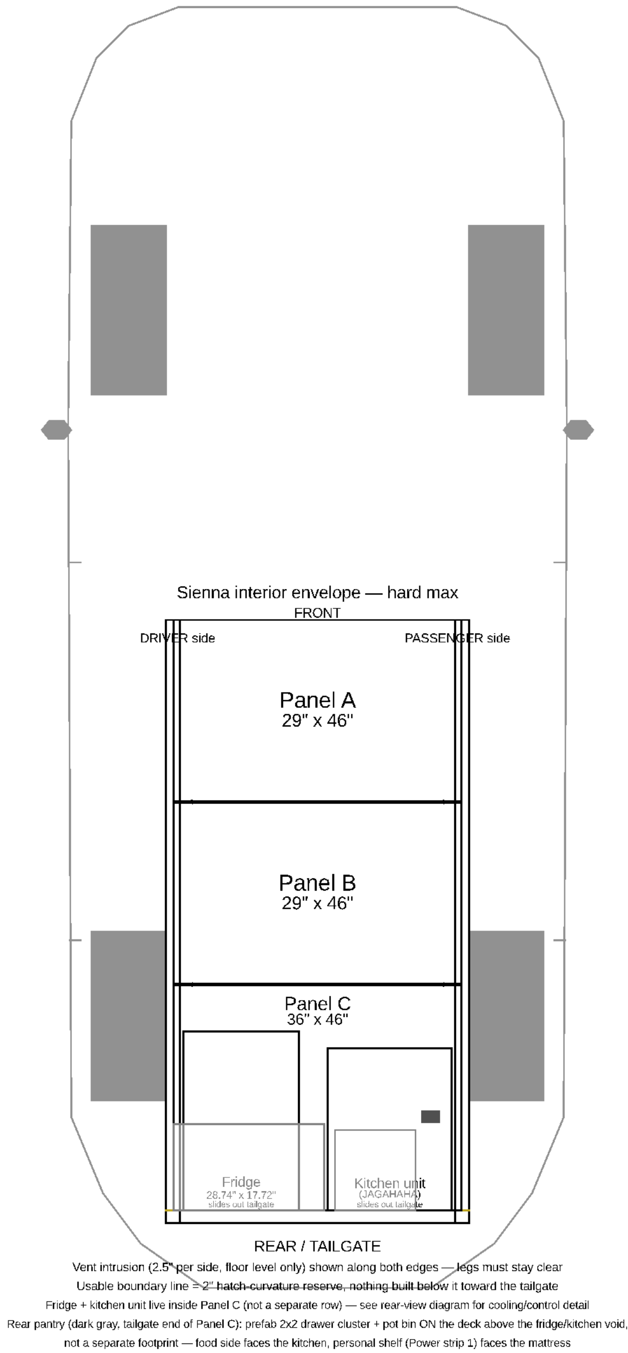
> **How to give feedback:** skim the seven drawings and the feature list, then jump to Open questions — those are the calls I'd most like a second opinion on (and the measurements still unverified). Tell me what's missing, what you'd do differently, or where the design looks fragile.

SLEEPS **2** 52"-wide platform, 25"/person	VAN INTERIOR **96 × 48.5 × 44"** L × W × H (length est.)	MODULES **3 + 4** Panels A/B/C + prefab drawer pantry	POWER **DELTA 3 +** Extra Battery · WAVE 3 A/C
COLD **41 QT** Rocky 40 dual-zone, slide-out	EST. COST **$3.1–3.5k** full; ~$2.4k if gear owned		

THE LAYOUT

One continuous deck of three lift-out boxes; the fridge, kitchen, and pantry all share Panel C at the tailgate end.

BIRD'S-EYE FLOOR PLAN

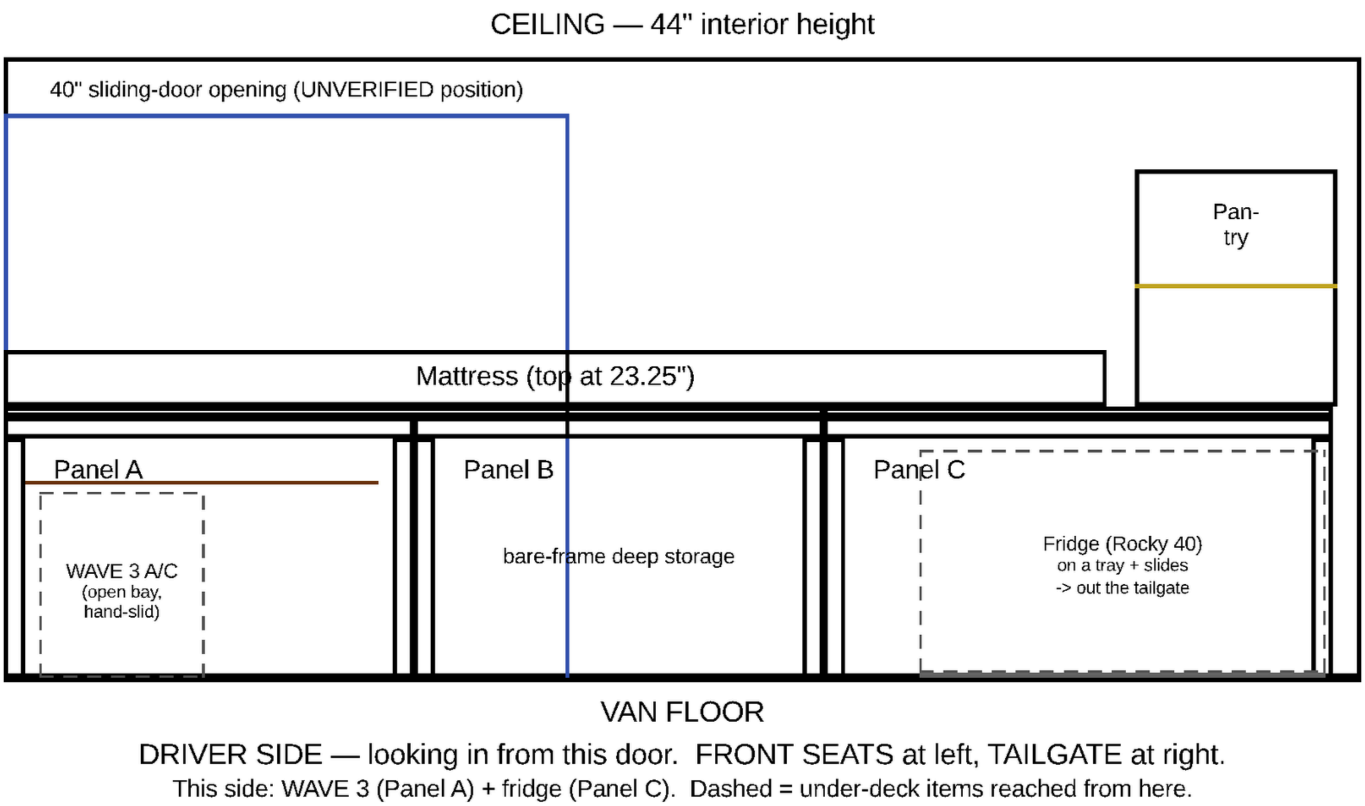


Top-down. Front of the van at top, tailgate at bottom. Panel A (29") + Panel B (29") carry the bed; Panel C (36") holds the fridge (driver side) and kitchen (passenger side) underneath, with the prefab drawer pantry on the deck above them. Grey blocks = wheel wells; the platform stays clear of the floor-level heat vents at the edges.

THE DRAWINGS

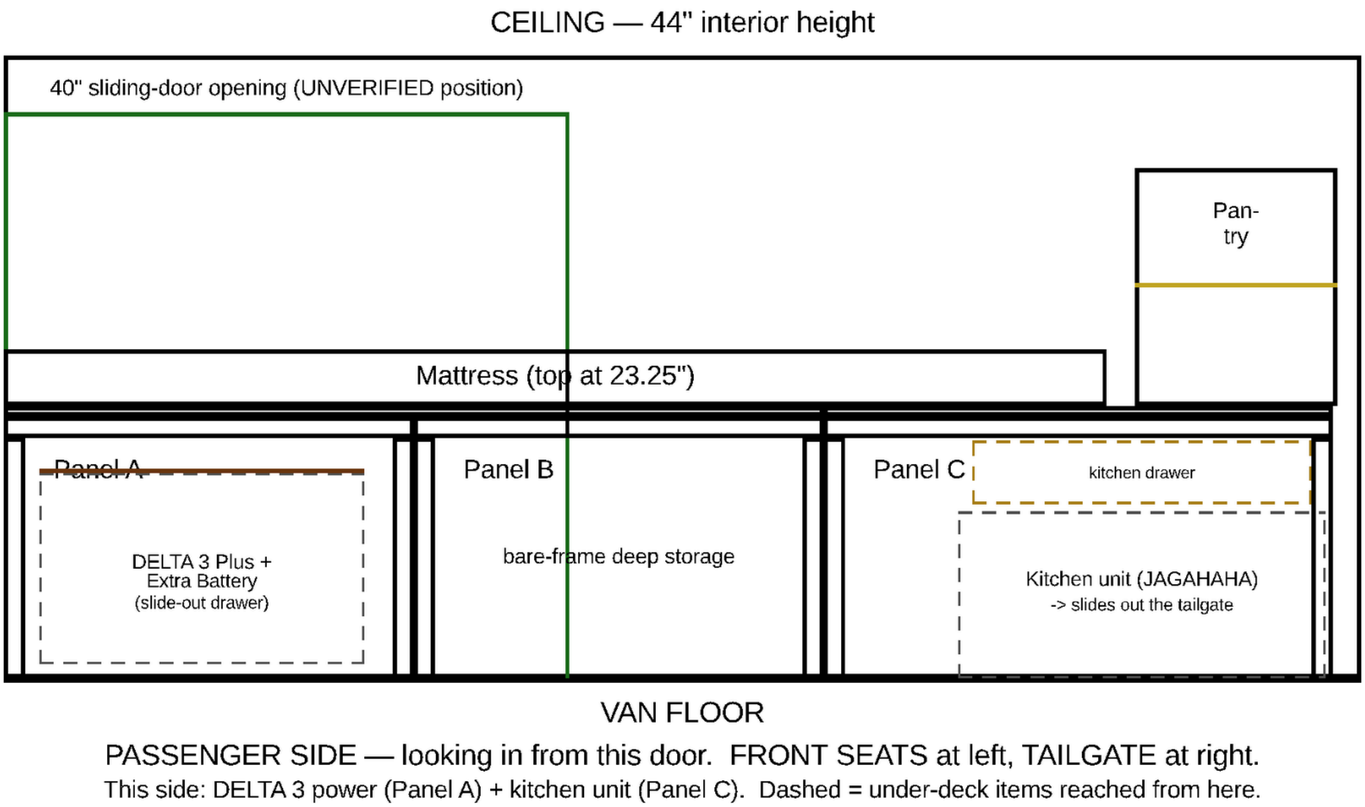
The two side elevations show what each sliding door reaches; the two gate views show the galley stowed for driving vs. deployed for use.

DRIVER SIDE



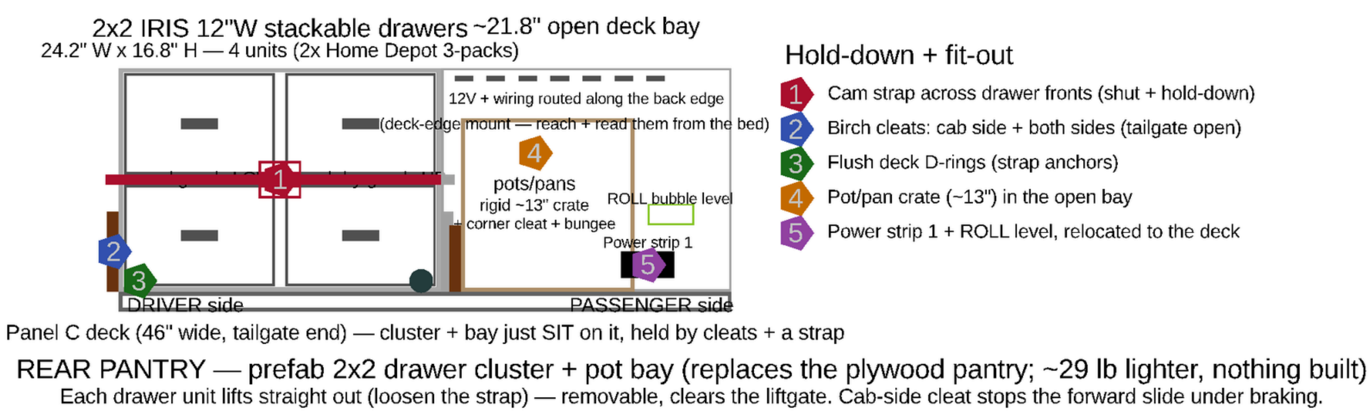
From the driver door. WAVE 3 A/C in Panel A's open bay; the fridge under Panel C. Dashed = under-deck items reached from this side.

PASSENGER SIDE



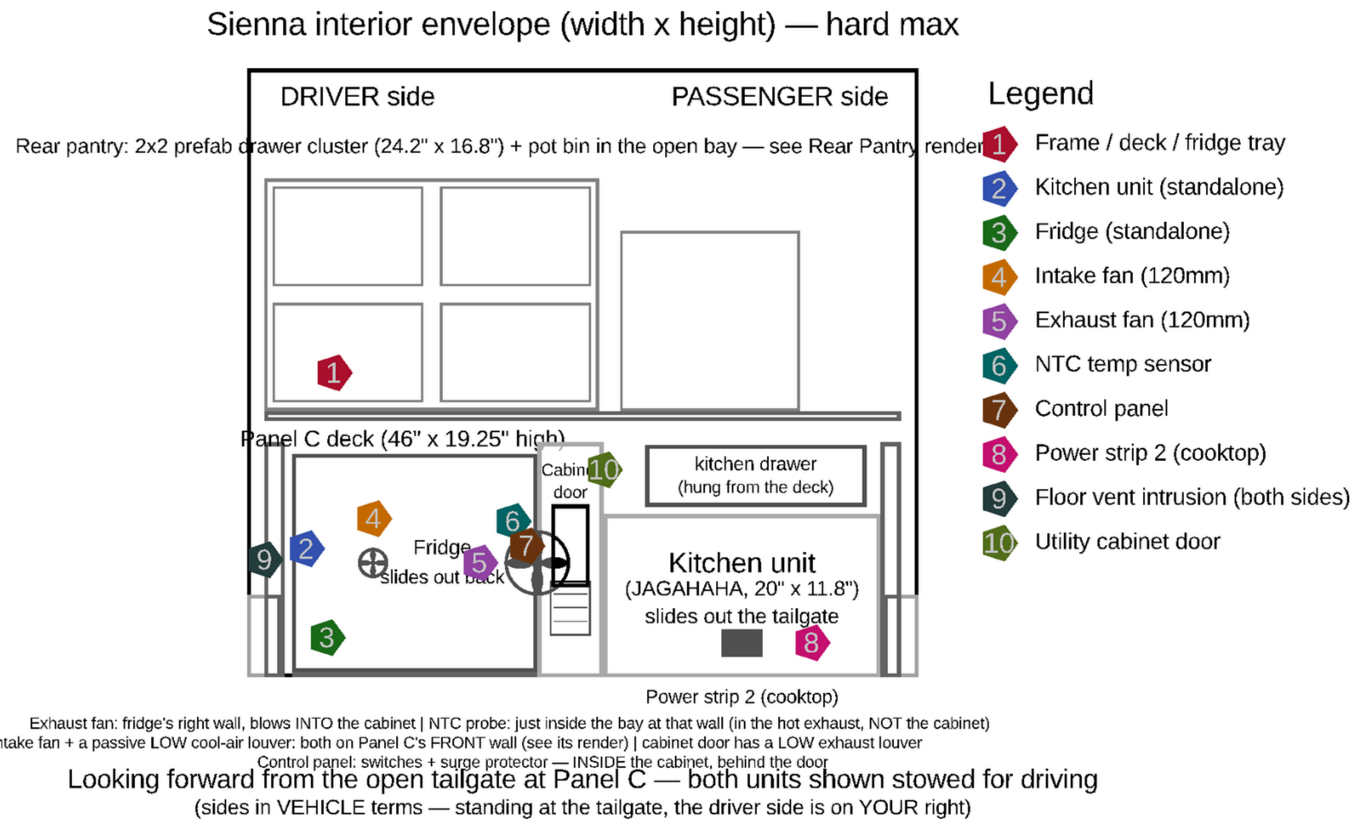
From the passenger door. The DELTA 3 power drawer in Panel A; the kitchen unit + its hung drawer under Panel C.

REAR PANTRY — PREFAB DRAWER CLUSTER



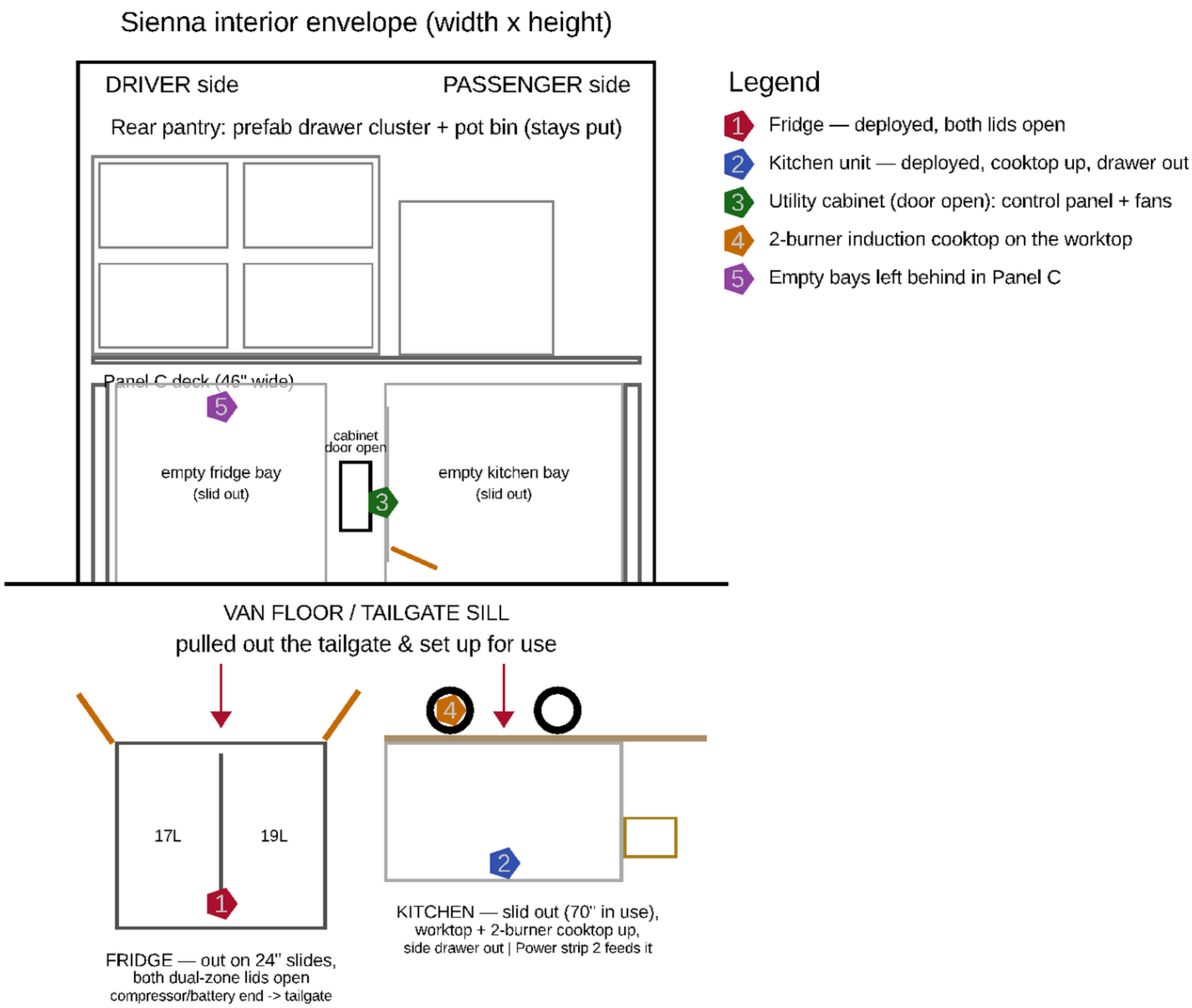
Bought, not built. The pantry is a 2×2 stack of IRIS 12"W drawers (24.2" × 14.3" × 16.8", ~15 lb) on Panel C's deck, plus a ~13" pot crate in the leftover 21.8" bay. A cleat pocket + one cam strap hold everything; each drawer unit lifts straight out. Power strip 1 + the roll bubble level mount on the deck edge beside it.

INTO THE GATE — STOWED



Driving mode. Fridge + kitchen tucked under Panel C's deck; cooling fans, NTC sensor and control panel live in the utility cabinet between them.

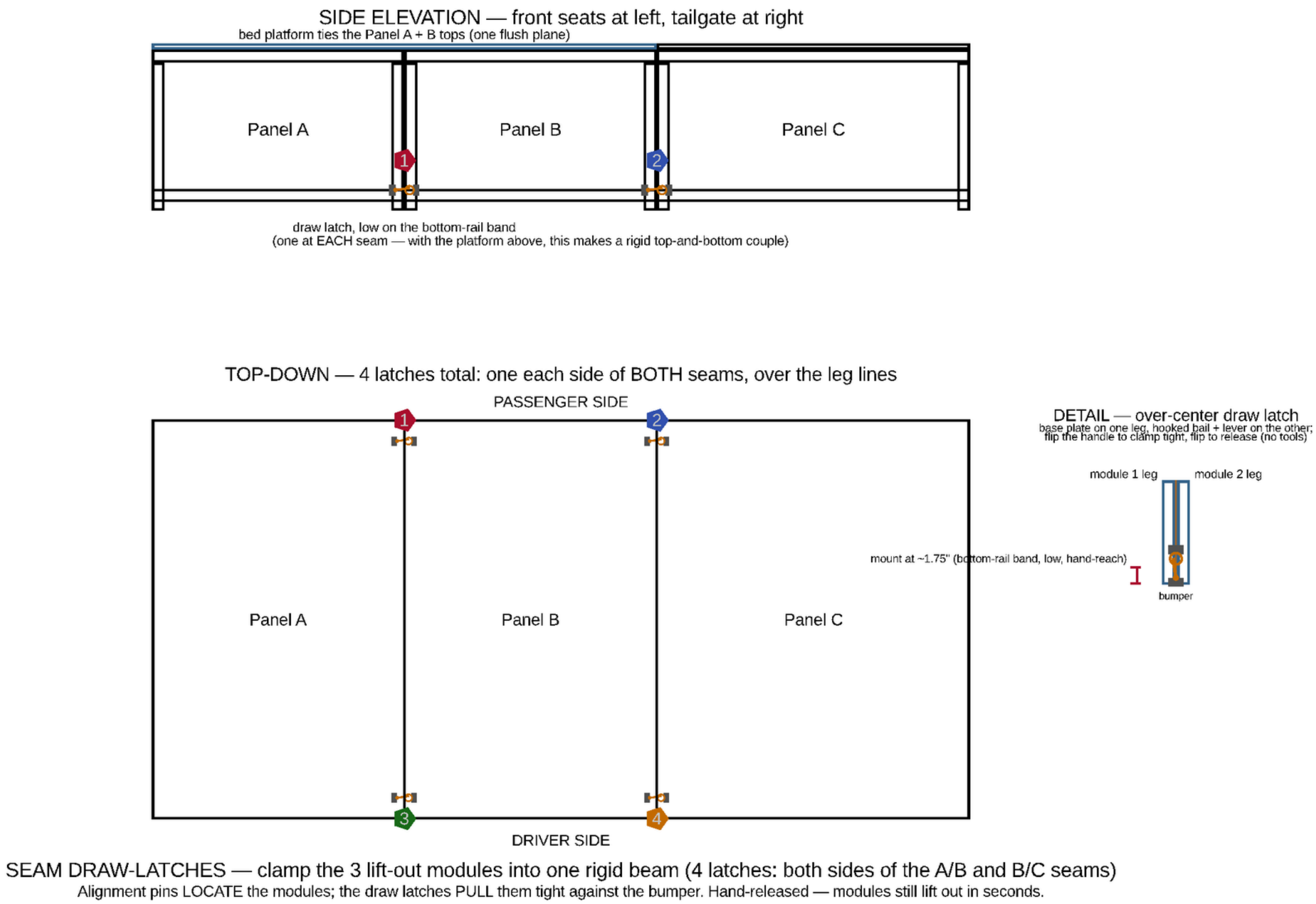
INTO THE GATE — DEPLOYED



DEPLOYED — looking into the open tailgate with the fridge & kitchen pulled out and set up for use

Camp mode. Both slide out the tailgate: fridge lids open (dual-zone 17L+19L), kitchen worktop + 2-burner cooktop up and drawer out. The bays behind them sit empty.

SEAM DRAW-LATCH POSITIONING



Clamping the modules. The alignment pins only *locate* the three boxes; four hand-released over-center draw-latches — one on each side of both seams, low on the bottom-rail band — pull them tight into one rigid beam. With the bed platform tying the tops, the low latches complete a top-and-bottom couple that kills sway and rattle, while any panel still lifts out in seconds.

FEATURES

What each part is and why it's there. Amber notes flag the ones I'd most like feedback on.

STRUCTURE
Modular lift-out platform

Three independent boxes — Panel A (29"), B (29"), C (36"), all 46" wide — built on 2×2 pine frames with leveling legs, plus a clamp-on pantry. Nothing bolts to the van; each box lifts out through the liftgate as a one/two-person carry. Cube-frame bottom rails add racking stiffness, and **4 hand-released over-center draw-latches** at the seams clamp the three boxes into one rigid beam without giving up the lift-out.

Built weight is **≈ 405 lb** (after lighter-wood swaps), ~884 lb loaded with two people — comfortably under the van's payload, though the mass is rear-biased (see the weight budget in the plan).

SLEEPING
Cantilevered bed platform

A 52"-wide slatted 1×4 platform spans Panels A+B and ends flush at the B/C seam, meeting Panel C's own deck. It overhangs the 46" boxes by 3" per side — using the fact that the van is wider up at mattress height than at the pinched floor.

Feedback wanted — this depends on ≥53" of wall-to-wall width at 19–23" above the floor, which isn't measured yet (see question 3).

SLEEPING
Mattress

HEST Dually Long — 78 × 50 × 4" solid foam with a washable waterproof cover, 25" per person. Leaves roughly 16.75" of sitting headroom to the ceiling.

GALLEY
Rear pantry — prefab drawers

Storage is bought, not built: four IRIS USA 12"W stackable drawers (Home Depot 3-packs) stacked 2×2 on Panel C's deck at the tailgate — 24.2" wide, 16.8" tall, ~15 lb — plus a rigid pot crate in the remaining ~22" of deck, and 4 Sterilite 28-Qt lidded totes containerizing Panel B's top-load bay. A screwed-down cleat pocket stops it sliding under braking, and one cam-buckle strap across the drawer fronts keeps the drawers shut and the stack snug; loosen it and any unit lifts out (~3 lb each). No joinery, no retention hardware, and the drawers contain their own contents.

Feedback wanted — would you trust fabric/plastic prefab drawers + a strap on washboard roads, or is the strapped-down rigidity worth checking on a rough test drive first?

`GALLEY` **Fridge**

BougeRV Rocky 40 — dual-zone 41 QT, 40.6 lb, 60W max — on a heavy-duty 24" slide + plywood tray, pulled out the open tailgate for lid access. The compressor/battery end faces the tailgate so the optional battery swaps without a full slide-out.

`GALLEY` **Kitchen unit**

JAGAHAHA slide-out camp kitchen (2-burner, own drawer), strapped to floor anchors, sliding out the tailgate and extending to ~70" in use. A shallow kitchen drawer hangs in the dead air above it for utensils and flat goods. The induction cooktop runs off Power strip 2.

`COOLING` **Forced-air fridge bay**

Two 120mm fans — intake on Panel C's front wall, exhaust on the fridge's kitchen-facing wall blowing into the utility cabinet — on a W1209 thermostat reading an NTC probe in the hot exhaust, so they only spin when the compressor is working. Two passive louvers (low intake + cabinet exhaust) back them up. This compensates for the bay being tighter than the fridge's passive-venting spec.

`POWER` **EcoFlow system**

DELTA 3 Plus + Smart Extra Battery ride in Panel A's slide-out drawer (charged at camp via shore power/solar, not from the van). The WAVE 3 air conditioner stores in Panel A's open bay and runs at the tailgate or the front seat. Two AC power strips feed the cooktop and the bed cubby.

`LEVELING` **Two-stage leveling**

Level the van at the wheels with Lynx blocks, guided by a phone Block Calculator that reads two bed-mounted bubble levels and tells you blocks-per-tire. The interior leg feet are then set once against the van's own floor. (Electric feet were considered and rejected.)

STORAGE

Found space + joinery

Reclaimed dead headroom — a tray over the DELTA 3 stack, a shelf over the WAVE 3, bins in the cabinet gap — and Panel B is a bare-frame deep-storage cube. The bottom stabilization frame was dropped lower for more usable volume. Carcasses use Ryobi DBJ50 biscuit joints (R-series) for strength and clean faces, with fastener locations dimensioned.

SAFETY

Air & alarms

An owner-placed low-mounted CO monitor (audible from the bed) and a fire extinguisher, plus SRS seat-delete emulators so the airbag system reads normally with the 2nd row removed.

OPEN QUESTIONS FOR REVIEWERS

The unverified measurements and judgment calls. Items tagged VERIFY need a tape measure on the actual van before anything is cut.

1 **Interior length** VERIFY

96" is an estimate for the empty interior with the 2nd row removed. Measure closed-hatch to front seatbacks — the whole assembly is sized to fill it exactly.

2 **Liftgate opening** VERIFY

48 × 36" is web-sourced, not measured. Every module must pass through it upright, and the upper corners round off — measure the real narrowest width and height.

3 **Bed cantilever width** VERIFY

The 52" platform overhangs the 46" boxes and needs ≥53" wall-to-wall at 19–23" above the floor along the whole run. If it's tighter, the platform shrinks.

4 **Side-door position & reach** VERIFY

The door's fore-aft position is a placeholder. It decides which drawers are actually reachable from the side vs. blocked by body structure.

5 **Rear-corner floor vents** VERIFY

Panel C's rear legs sit at the true corners; confirm the floor heat vents don't reach the last ~4" there.

6 **Pantry cluster vs. liftgate swing** VERIFY

The 18.6" drawer cluster + pot bin sit at the tailgate. Confirm the gate's swing-up clears them and the fridge/kitchen still slide out past.

7 **Appliance specs & price** VERIFY

Confirm live-listing dimensions and current price for the Rocky 40 fridge and the JAGAHAHA kitchen before ordering.

8 **Floor anchors**

The E-track anchors bolt through the floor pan — inspect underneath for fuel/brake lines and wiring, and seal each hole.

DESIGN CALLS WORTH A SECOND OPINION

Beyond the measurements, these are open on the merits — I'd value a view:

- Is a **slide-out kitchen + separate fridge** the right split, or would one integrated galley be simpler and stronger?
- Is **two 120mm fans + passive louvers** enough fridge cooling for real summer heat, or should the bay get a bigger fan or a dedicated vent?
- Prefab drawer pantry at the tailgate vs. moving it forward — does having it right at the gate get in the way in practice?
- Anything **missing entirely** — water/sink, interior lighting, a shore-power inlet, insulation, screens?

COMPANION TOOLS

The interactive/printable sheets that go with this plan.

Block Calculator →
Blocks-per-tire from the bubble levels

Shopping List →
Materials, quantities & links

Cut List →
Every board & panel, grouped

Build Sequence →
Dependency-ordered assembly

Drawings are to-scale OpenSCAD line art from the parametric model; every dimension is computed from the plan's source of truth, not sketched. Full detail lives in **Project_Smores.pdf** (Components 1–10 + appendices). Estimates and unverified figures are flagged throughout.